Predictive Analysis for Socioeconomic Status

AJ Savarese and Artemas Souder

Kennesaw State University

DATA 4310: Statistical Data Mining

Dr. Duleep Rathgamage Don

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Abstract

This study looks at how socioeconomic status (SES) can help predict if a certain tract in Georgia is wealthy or struggling. We investigated how one's cost of living changes depending on a neighborhood's socioeconomic status. We used data from the American Community Survey for 2,796 areas in Georgia. Missing values and outliers were addressed, and many different models were created to visualize our variables of interest. New variables were created to test SES, one based on the formula, and one through creating bins to see the range of SES. We tested many different models to see which ones could best predict these status levels. These models include Naïve Bayes, Random Forest, K-Nearest Neighbor, and Multilayer Perceptron. The accuracy scores were quite close, ranging from a low of 0.69 to a high of 0.72. The MLP model was the most accurate, followed by Random Forest, Naïve Bayes, and finally KNN. The most important clues for the models were home values and rent prices. These results show that the cost of housing is the biggest indicator for predicting a community's socioeconomic level.

Introduction

Research Problem Definition / Motivation

Understanding socioeconomic status (SES) is very important when looking at issues like income inequality and housing poverty. In Georgia, any disparities in wealth and education can significantly impact the cost of living and overall quality of life. It is also important to understand what the major issues are, and hopefully better address them for the wellbeing of many people.

Importance of Classification

Classification is appropriate for this research because we need to be able to categorize a bunch of complex census data into an SES ranking that we can better understand.

Objective Definition

We wanted to represent SES as a variable that we could then use to test our hypotheses and come to any conclusions based on different demographic predictors.

Target Variable

Our target variable is Socioeconomic Status (SES), which we created based on a formula as our way to represent opportunity level for a tract.

Research Question

We had two main questions to push our research. One was seeing how demographic factors predict a county's socioeconomic status (SES). Then, diving deeper with that, we were also curious to see how cost of living might differ for different socioeconomic statuses (SES).

Data Description

Dataset Overview

Our dataset is the American Community Survey (ACS) 5-Year estimates at the Census tract level for Georgia (2024). There are 2796 observations (census tracts) with 19 different variables.

Target Variable Construction

Our target is SES_bin, a categorical variable created to group SES into 5 separate groups, ranging from 'Very Low' to 'Very High'. We also created a variable for SES (SES_index) that we used to create initial values.

Feature Descriptions

Because we are calculating more on a neighborhood level, our SES formula follows the standard metrics used by organizations like the U.S. Census Bureau to identify community needs.

Missing Value Handling

The missing values we focused on were population, labor, income, and race. The reason for this is because those zero values are literally impossible to have in a tract. We removed these inputs from the dataset, but kept other variables' zeroes we deemed technically possible.

Outlier Detection

We mainly determined the outliers through visual representation. We did run IQR, and most seemed within reason; the only extremes to note were instead caused by missing values (-666666666). Those were removed accordingly.

Feature Transformations

We used standardization to scale our values, as we're working with different ranges of variables and wanted to avoid any statistical bias. We also did log transformations to handle any skewed values.

Exploratory Data Analysis (EDA)

Univariate Analysis

To begin our exploratory data analysis, we completed a univariate analysis using descriptive statistics, histograms [Fig. 1], skew, and boxplots [Fig. 2]. We created our target variable socioeconomic status (SES) [Fig 3 & 4] where observations were placed into five categories [Table 1]. Many of the variables have a right-skewed distribution, with them all

having a number of outliers. However, age and SES were the variable closest to a normal distribution.

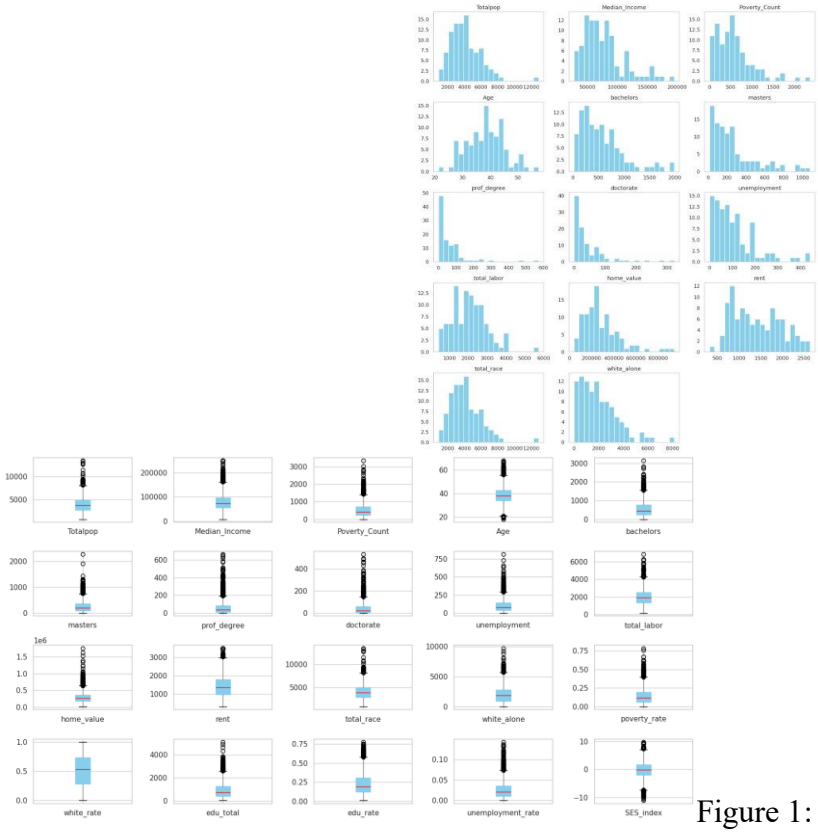


Figure 1: Univariate Analysis Histograms

Figure 2: Univariate Analysis Boxplots

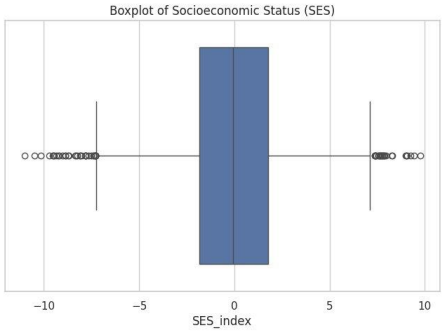


Figure 3: SES Creation Boxplot

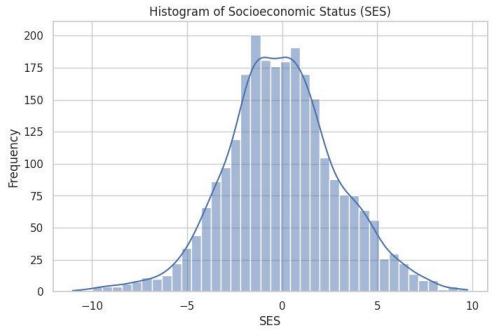


Figure 4: SES Creation Histogram

SES Bin	(N)
Very Low SES	40
Low SES	372
Medium SES	1388

High SES	641
Very High SES	92

Table 1: SES Bins

Bivariate Analysis

EDA continued through bivariate analysis with a look at the boxplots for each variable based on SES categories [Fig. 5] and the scatterplots [Fig. 6] between each feature. The bivariate analysis showed differences between SES categories.

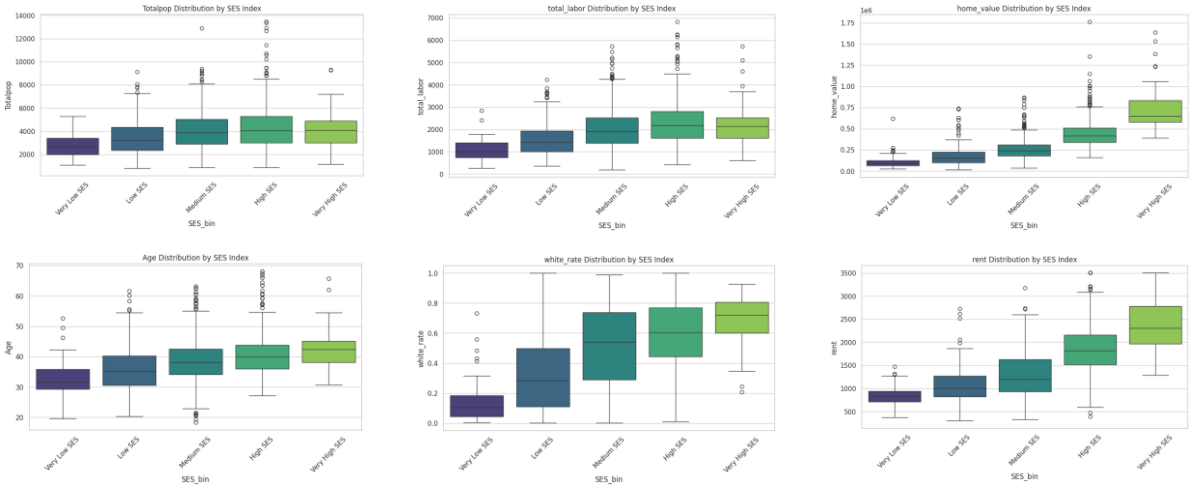


Figure 5: SES Bins Bivariate Boxplots

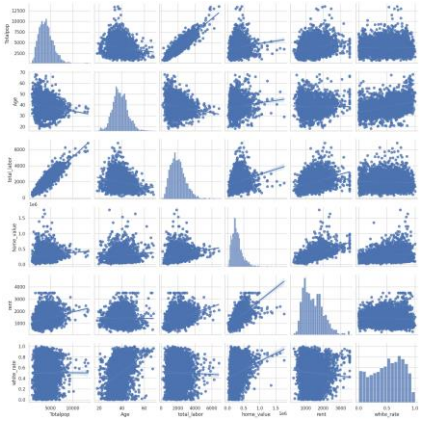


Figure 6: Pair Plot Scatterplots

Multivariate Analysis

Finally, EDA continued through comparison of mutual information scores [Fig. 7] and power transformation. Further analysis was completed using a correlation matrix [Fig. 8] with

rent and home value along with total labor and total population being the only variables with high correlation.

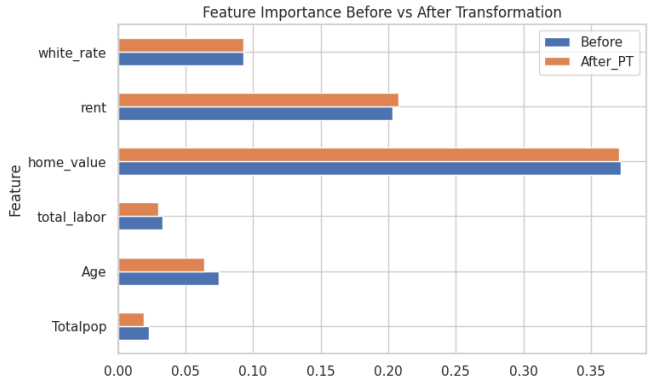


Figure 7: Feature Importance Plot

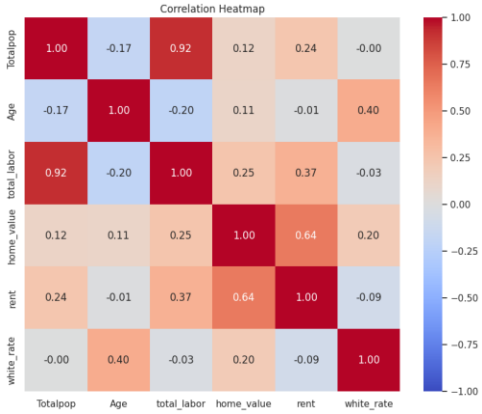


Figure 8: Correlational Heatmap

Key Insights from EDA

From the EDA we were able to conclude that median home value and median rent were the most important features. For the highly correlated variables, total population was removed from model analysis while home value and rent remained because of their different populations. When looking at the relationship between variables, most showed little to no linear relationship.

Model Building

Model Selection

We decided on completing 4 models: Random Forest, Naïve Bayes, K-Nearest Neighbor (KNN), and Multilayer Perceptron (MLP).

Model Formulation

The features used for the model were total labor force, median rent, median home value, age, and percent of white population. Assumptions for all models were met. For Random Forest, due to the differences in renting and home value populations, the independence assumption was affirmed. Due to the central limit theorem, we can assume normality for Naïve Bayes [Fig. 9]. Next, KNN made use of Euclidean distance even with the preference for smaller sample sizes. Finally, there was the MLP [Fig. 10] where all assumptions were met.

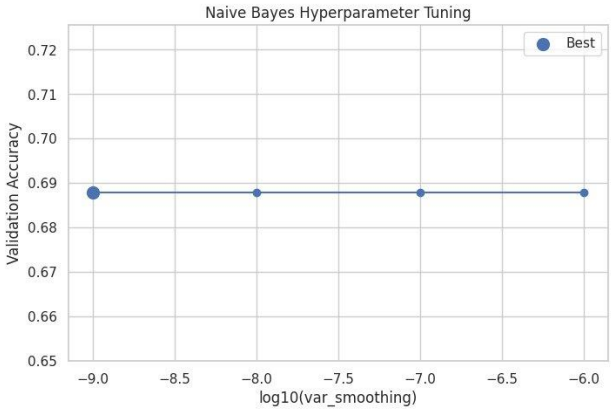


Figure 9: Naive Bayes Hyper Parameter Tuning

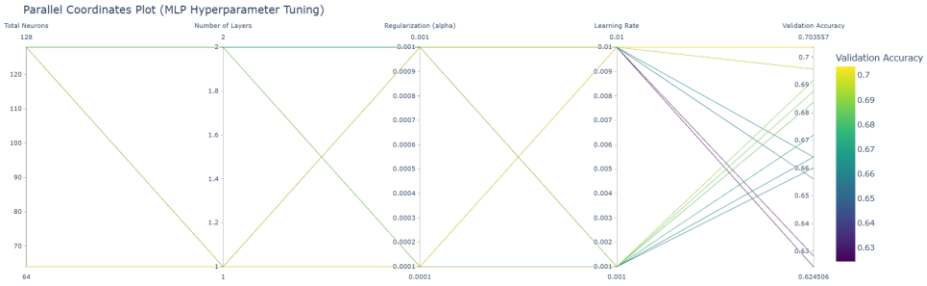


Figure 10: MLP Model Creation

Training Procedure

To build the model, we split the dataset using 70% for training and 30% for testing the model. We also explored which values would prove to be the most effective for each model.

Evaluation Metrics

Finally, to evaluate the model, we completed a confusion matrix along with calculating the accuracy, precision, recall, and F1 scores for the models. The accuracies of the models were as follows [Table 2]: MLP (.72), Random Forest (.70), Naïve Bayes (.69), and KNN model (.68). With the most important feature for each model being home value and the least important feature being median age.

Naive Bayes
Accuracy: 0.6923076923076923

Classification Report:

	precision	recall	f1-score	support
0	0.30	0.38	0.33	8
1	0.60	0.41	0.49	75
2	0.76	0.79	0.77	277
3	0.62	0.70	0.66	128
4	0.73	0.42	0.53	19
accuracy			0.69	507
macro avg	0.60	0.54	0.56	507
weighted avg	0.69	0.69	0.69	507

Confusion Matrix:

```
[[  3   5   0   0   0]
 [  7  31  36   1   0]
 [  0  16 219  42   0]
 [  0   0  35  90   3]
 [  0   0   0  11   8]]
```

K-Nearest Neighbor
Accuracy: 0.6804733727810651

Classification Report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	8
1	0.60	0.20	0.30	75
2	0.69	0.92	0.79	277
3	0.67	0.59	0.62	128
4	0.00	0.00	0.00	19
accuracy			0.68	507
macro avg	0.39	0.34	0.34	507
weighted avg	0.63	0.68	0.63	507

Confusion Matrix:

```
[[  0   6   2   0   0]
 [  0  15  60   0   0]
 [  0   4 255  18   0]
 [  0   0  53  75   0]
 [  0   0   0  19   0]]
```

Random Forest
Accuracy: 0.7041420118343196

Classification Report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	8
1	0.69	0.32	0.44	75
2	0.72	0.88	0.79	277
3	0.67	0.64	0.66	128
4	0.67	0.32	0.43	19
accuracy			0.70	507
macro avg	0.55	0.43	0.46	507
weighted avg	0.69	0.70	0.68	507

Confusion Matrix:

```
[[  0   6   2   0   0]
 [  0  24  51   0   0]
 [  0   5 245  27   0]
 [  0   0  43  82   3]
 [  0   0   0  13   6]]
```

MLP
Accuracy: 0.7199211045364892

Classification Report:

	precision	recall	f1-score	support
0	0.25	0.25	0.25	8
1	0.67	0.48	0.56	75
2	0.75	0.87	0.81	277
3	0.70	0.61	0.65	128
4	0.64	0.47	0.55	19
accuracy			0.72	507
macro avg	0.60	0.54	0.56	507
weighted avg	0.71	0.72	0.71	507

Confusion Matrix:

```
[[  2   5   1   0   0]
 [  5  36  34   0   0]
 [  1  12 240  24   0]
 [  0   1  44  78   5]
 [  0   0   0  10   9]]
```

Table 2: Model Classification Reports

Results and Discussion

Key Findings

The key findings for our models were that each model was decently important, with the lowest having an accuracy of .69 and the highest having an accuracy of .72. These indicate that the models were relatively similar, with differences being seen by differences in complexity of the models. The important predictors seen within all models were median home value and median rent price, while the least important features were age and percentage of the white population in that region.

Comparison with EDA

Our results prove to be very similar to what we expected from the EDA, with home value and rent continuing to be very important features for predicting SES. However, age proved to be very unimportant as a feature for determining SES. Many of the other features seemed to remain very similar. With the EDA we were able to create a model that was impactful and relevant to different populations within Georgia.

Limitations

The limitations found for this research included the need to have weights for different regions. Furthermore, there were some concerns about the bias of the models based on population density. Further limitations can be seen in our lack of analysis for the race distributions for each region.

Conclusion

To summarize our findings, we found that the MLP model was the best model, with median home value being the most important feature for determining socioeconomic status. The model's strength in order was MLP, Random Forest, Naïve Bayes, and finally, KNN. We were able to create these models that were adequate at predicting the counties' socioeconomic status in comparison to other counties. These findings are heavily influenced by the urbanization of the county. Furthermore, the models are heavily biased towards regions with larger urban centers. The practical implications of our model would be to advise and explain where targeted efforts to assist communities should be directed. Future research should be directed towards exploring this analysis from a city-to-city basis weighted by population density.

References

US Census Bureau. “How the Census Bureau Measures Poverty.” *Census.gov*, US Census Bureau, 9 Apr. 2025, www.census.gov/topics/income-poverty/poverty/guidance/poverty-measures.html.